A Stochastic Inventory Routing Problem for Infectious Medical Waste Collection

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We consider the problem of designing a logistic system to organize adequate collection of infectious medical waste. Waste materials are produced by patients in self-treatment, stored at pharmacies and picked up by local authorities for disposal. The problem is formulated as a collector-managed inventory routing problem, encompassing stochastic aspects, which are common in such problems. Social objectives, specifically the satisfaction of pharmacists and local authorities, as well as the minimization of public health risks, are considered for the real-world-motivated inventory routing problem. To optimize the planning process for a predefined time horizon, we take advantage of radio frequency identification technology, which allows an improvement of the planning process for pharmacists as well as local authorities. We develop two solution approaches to optimize the determination of visit dates and corresponding vehicle routes. We design a sampling method and an approach based on an adaptive large neighborhood search algorithm to treat the stochastic problem. The suggested approaches are tested on real-world data from the region of Provence-Alpes-Côte d'Azur, in France. To evaluate the performance of the proposed solution methods, the results for different waste collection scenarios are analyzed and compared. © 2013 Wiley Periodicals, Inc. NETWORKS, Vol. 63(1), 82–95 2014

Keywords: medical waste collection; stochastic inventory routing; adaptive large neighborhood search; sustainable logistics

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1. INTRODUCTION

This study is part of a project called *Trace-de-TIC*, funded by the French Environment and Energy Management Agency (ADEME). It is also connected to the *PAC ID DASRI* project, funded by the Provence-Alpes-Côte d'Azur (PACA) region in France and the European Union. These two projects notably investigate the use of radio frequency identification (RFID) technologies for infectious waste management.

Infectious waste materials considered in these projects relate to waste materials disposed of by home health and medical care services, as well as by self-attending patients. Most of them, especially sharp objects such as contaminated needles or syringes, need to be handled with great care to prevent the spread of diseases. In view of the development of home care services and the aging of the population, one can observe a fast increase in these types of waste [22]. There is therefore a growing importance in securing the complete disposal process: the unsafe disposal of health-care waste indeed poses major public health risks, considering that health workers, waste handlers, or pedestrians might come into contact with contaminated waste if it is dumped into areas without restricted access. As a matter of fact, the world health organization recently established strict guidelines for the management of infectious waste materials [33].

The purpose of the *DASRI* project was to improve the traceability in the treatment of infectious waste, by developing automated RFID containers (Fig. 1b). With these containers, located in pharmacies, dedicated boxes containing infectious waste (Fig. 1a) can be identified and traced until the final stage of the treatment. Within the *Trace-de-TIC* project, and in this article, we investigate how new information provided by RFID can also help improving the planning of the disposals.

In the south-east region of France, the disposal of medical waste accumulated by patients in self-treatment is commonly



(a) Medical waste box (b) Medical waste container

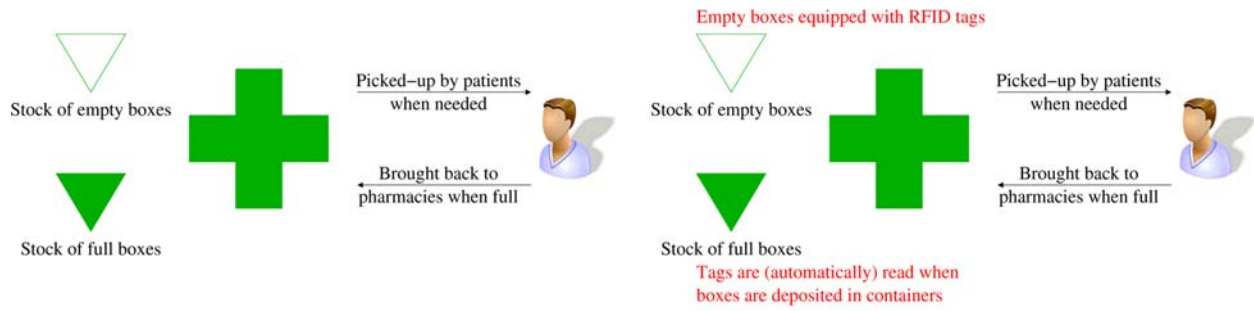
FIG. 1. Infectious medical waste containers. [Color figure can be viewed in the online issue, which is available at wileyonlinelibray.com.]

organized as follows (Fig. 2a). At pharmacies, patients receive empty boxes which are dedicated to the safe storage of their medical waste. Full boxes are brought back by the patients to the pharmacies, where they are stored. When a certain storage capacity at a pharmacy is reached, a pharmacist makes a call to the local authority, which organizes the pick-up of the waste material. Full boxes are then collected by a vehicle and deposited at collection centers for disposal. An external company periodically visits and depletes collection centers.

Currently, a pharmacist makes a call to the local authority when the maximal storage capacity (a container for full boxes) at the pharmacy is reached. The pharmacy is then visited in one of the two following days. The current system has three main inconveniences. First, transportation costs are unduly high: as the number of boxes at pharmacies is only known by the collector when a call is received and as pharmacists do not expect any visit before calling the collector, no anticipation is possible. Hence, no optimization based on geographical clustering can be introduced when designing collection routes. Second, the service at pharmacies is bad, not only for pharmacists but also for patients, and might raise some risks: if a container is full, waiting to be emptied by the collector, patients must have recourse to the pharmacist for finding another storage place, causing some disturbance for both of them and possibly some danger depending on the available space. Third, no clear vision on future collection tasks is given to the collector. As this activity typically only represents a limited part of his/her daily work, managing new collection requests complicates the rest of his/her work.

Being equipped with RFID tags (Fig. 2b), boxes can automatically be registered when deposited at a pharmacy without disturbing the pharmacist. This provides two valuable pieces of information to the collector. First, with RFID, the exact number of boxes in containers can be accessed at any time. Second, statistics on full-box arrival rates can be recorded. Based on this information, a new collection policy can be carried out, with waste collection tours planned *a priori* at the beginning of the planning horizon for the whole period, depending on the actual number of boxes, the forecasted arrival rates and the distances between pharmacies. We call this policy the RFID scenario. Basically, this policy should help limiting the three inconveniences of the current process. The flexibility in fixing the visit dates of pharmacies makes the simultaneous optimization of collection days and routing possible. Having statistics on arrival rates and entire freedom to plan collections should permit avoiding most of the stock excesses. Possessing information on when collections of waste material will be carried out during the whole planning period, collectors can adjust their time schedule and better organize their daily activities. Nevertheless, the whole process becomes less reactive as the planning is based on information available at the beginning of the time horizon. However, real-time information on inventory levels at each pharmacy can be used each day to revise the planning of collection tours. This allows another collection policy to be carried out by revising solutions found by the RFID scenario each day based on actual arrivals of data. We call this policy Rolling horizon scenario. From a methodological point of view, the Rolling horizon scenario does not add any complexity to the problem, as it is solved with the tools which have been developed for the static case (RFID scenario). Therefore, in the course of the article, we focus on the static RFID scenario. Details concerning the Rolling horizon scenario follow when computational experiments are presented.

Two conflicting aspects have to be considered when determining visit dates for pharmacies. On the one hand, pharmacies should not be visited too late, as their storage



(a) Current process (b) Current process with RFID

FIG. 2. Medical waste collection process. [Color figure can be viewed in the online issue, which is available at wileyonlinelibray.com.]

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capacity will be exceeded, thus rising some risk and disturbance as explained before. On the other hand, pharmacies should not be visited too early (when only few boxes are stored), as each visit disturbs people present in the pharmacy (including the pharmacists).

Our study contains several innovative and new aspects. First of all, the problem formulation combines characteristics of inventory routing and stochasticity in an applied research study. Second, we consider a mix of four economic and social performance indicators for the collection of infectious medical waste, namely minimization of public health risks, satisfaction of pharmacists, satisfaction of the local authority, and monetary cost of collection tours. Then, we provide a mathematical formulation of the waste collection problem described above, addressed as a stochastic inventory routing problem (IRP). Finally, we develop several heuristic solution methods, that we compare, and we analyze the performance of the new policies against the actual one on real-world data from the PACA region.

In Nolz et al. [24], the infectious medical waste collection problem was presented and a tabu search algorithm with preliminary results was detailed. In this article, we develop two additional solution approaches, based on sampling and on adaptive large neighborhood search (ALNS), respectively, and present extensive computational experiments.

The remainder of the article is organized as follows. In the next section, a literature review is presented. In section 3, the problem is described in detail and a mathematical formulation is provided. In section 4, solution approaches are introduced and explained. Different computational experiments are presented in section 5 to show the effectiveness of our solution approaches and discuss managerial implications of RFID. Concluding remarks are provided in section 6.

2. LITERATURE REVIEW

The infectious medical waste collection problem under investigation contains characteristics of inventory routing and stochasticity. It is furthermore embedded in an applied research project. This is why we partition the literature review into three sections: IRP, real-world waste collection problems, and stochastic vehicle routing problems.

2.1. Inventory Routing Problems

The IRP deals with the distribution (or collection) of a product to (from) several facilities over a given planning horizon (Bertazzi et al. [4]). In general, three different decisions are addressed: when to visit each pharmacy, how much to collect each time a pharmacy is served and how to route the vehicle(s). The general objective of an IRP is to minimize routing and inventory costs. For an overview on IRPs regarding different formulations and extensions, the reader is referred to Bertazzi et al. [4]. Federgruen and Simchi-Levi [13] provide a mathematical formulation of the single-depot single-period IRP. Campbell et al. [6] study the IRP as well as the stochastic IRP, where the future demand of a customer is uncertain. Stockout costs at a customer are modeled with a penalty function that contains fixed and variable components. Campbell and Hardin [8] study the problem of minimizing the number of vehicles for periodic deliveries, which is motivated by IRPs considering customer demands at a steady rate. Campbell and Savelsbergh [9] present a two-phase approach for an IRP. In the first phase, a delivery schedule is determined by solving an integer programming model and in the second phase a set of delivery routes is constructed by applying heuristics. Yu et al. [34] study an IRP with split deliveries, allowing the delivery to each customer in each period to be performed by multiple vehicles. Le Blanc et al. [19] present a collector-managed IRP for reverse logistics, where the collection company is responsible for the inventory to be picked up and recycled. Solyali et al. [29] present a robust IRP with uncertain demands. The authors determine the delivery quantities and delivery times, as well as the vehicle routes, and they allow backlogging of demand at customers. They propose mixed integer programming formulations of the problem and its deterministic variant.

2.2. Real-World Waste Collection Problems

Real-world waste collection problems have received a special interest in the last decade. Baptista et al. [3] present a periodic vehicle routing problem (PVRP) case study considering the collection of recycling paper containers in Portugal. The number of visits to containers within a period is included in the problem formulation as a decision variable. Nuortio et al. [25] address a stochastic PVRP with time windows for the collection of municipal solid waste in Eastern Finland, which is solved with a metaheuristic. Kulcar [17] presents a case study on waste collection in Brussels, evaluating several means of collection, such as vehicle, rail, and canal. Mourao and Almeida [23] examine a capacitated arc routing problem for a refuse collection VRP. They present lower bounding techniques and a heuristic to solve a real-world problem for the collection of household waste in Lisbon. Teixeira et al. [30] provide a case studied on the collection of recyclable urban waste. Periodic vehicle routes are planned for the collection of three different waste types, regarding a relatively long-time horizon. Tung and Pinnoi [32] studied a vehicle routing and scheduling problem for waste collection in Hannoi, Vietnam. The problem is divided into several phases, where waste has to be picked up and delivered at different stages, while time windows have to be respected. Alegre et al. [1] describe a PVRP for the pick-up of auto parts in northern Spain regarding a long-time horizon. The authors propose a metaheuristic solution approach based on scatter search. Campbell et al. [7] present a case study for a real-world IRP. They develop a two-phase solution approach. In the first phase, they determine the customers to be served on a specific day and the amount to be delivered, and in the second phase they determine the delivery routes and schedules. Shih and Chang [28] develop a computer system for the collection of infectious hospital waste. Their solution approach for the

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considered PVRP consists of two phases. First, the authors develop vehicle routes for all customers without considering allowable day combinations. In the second phase, they apply a mixed integer programming method to assign the routes to days while respecting all constraints. Faccio et al. [12] introduce a multiobjective vehicle routing model considering real time traceability data for the collection of solid waste in Italy.

2.3. Stochastic Vehicle Routing Problems

Gendreau et al. [14] provide a review paper on stochastic VRPs, referring to stochastic customers, stochastic travel times and stochastic demands. The same authors develop a Tabu search heuristic called Tabustoch [15] for the VRP with stochastic demands and customers. Laporte and Louveaux [18] present a two-index formulation of the SVRP and explain the integer L-shaped method for continuous stochastic programming, which is based on a branch-and-cut approach. Several papers focus on the solution of a stochastic program with recourse. In the first stage, an a priori solution is determined, where the realizations of the random variables are not known. In the second stage, a recourse or corrective action can be taken, if for example the vehicle capacity is exceeded. The aim is to determine a solution of minimum expected total costs. Tricoire et al. [31], for example, formulate a biobjective two-stage stochastic program with recourse for a covering tour problem with stochastic demand. They apply a branch-and-cut technique to a sample-average version of the problem within an epsilon constraint algorithm. Hjorring and Holt [16] use the L-shaped method to solve a single VRP. A relaxed IP is solved and additional constraints (optimality cuts) are added for a tighter approximation of route failure costs. Rei et al. [27] propose a hybrid heuristic combining Monte Carlo sampling and local branching to solve a single VRP with stochastic demands. Mendoza et al. [21] propose a memetic algorithm for the solution of a multicompartment VRP with stochastic demands. The authors apply genetic operators in combination with local search procedures to solve the stochastic program with recourse. Ak and Erera [2] present a paired-vehicle recourse strategy, considering the coordination of vehicles in pairs. A Tabu search heuristic is applied to real-world instances from Istanbul, Turkey. Christiansen and Lysgaard [10] introduce a branch-and-price algorithm for the capacitated VRP with stochastic demands. The authors formulate a set partitioning problem and develop a dynamic programming approach for the solution of the column generation subproblem.

3. PROBLEM DESCRIPTION

We coin our problem as the Stochastic collector managed IRP (SCMIRP). Two types of decisions have to be taken: (1) when to visit each pharmacy, (2) how to sequence vehicle routes.

The planning period is fixed to a given number of days (typically a few weeks). Based on the situation encountered in the PACA region, we consider a single vehicle for collection tours. The vehicle is located at the depot, where it starts and ends each of its tours. At most, one tour is allowed every day. The capacity of the vehicle is not considered, as being assumed to be sufficient to visit any possible subset of pharmacies. Arrival of new boxes in pharmacies is modeled using random variables, dependent on the pharmacy and the time period (day). The initial inventory level for each pharmacy is exactly known; no constraint is imposed on the inventory level at the end of the planning period.

In the following, the determination of visit dates and collection tours is performed once at the beginning of the planning horizon. This problem corresponds to the RFID scenario described in section 1.

Being equipped with RFID tags, boxes can automatically be registered when deposited at a pharmacy without disturbing the pharmacist. This provides two valuable pieces of information to the collector. First, with RFID, the exact number of boxes in containers can be accessed at any time. Second, statistics on full-box arrival rates can be recorded. Based on this information, a new collection policy can be carried out, with waste collection tours planned *a priori* at the beginning of the planning horizon for the whole period, depending on the actual number of boxes, the forecasted arrival rates, and the distances between pharmacies.

The objective function is composed of two types of costs: distribution and inventory. Distribution costs comprise routing costs and fixed costs incurred each time a tour is planned; the latter relate to fixed extra-times involved by each tour for the collector. Inventory costs reflect the aim of targeting a certain range in the number of boxes actually picked up by the vehicle when visiting a pharmacy. Unitary excess inventory costs are counted when the container maximum storage capacity (number of full boxes) at a pharmacy is exceeded; these costs reflect the perturbance and risk that every box in excess provokes. A penalty cost is incurred when the number of boxes at a pharmacy is lower than a predefined threshold inventory level; this cost reflects the difficulty for a pharmacist to accept a visit as long as a minimum storage level has not been reached, because the visit perturbates the normal functioning of the pharmacy. Regarding the final inventory level, unitary excess inventory costs are also considered, in the same fashion as when a visit is planned, so as to avoid delaying collection costs to future periods.

Note that the way inventory costs are modeled and the values given to the different parameters in these costs (for the PACA region) were defined in agreement with pharmacists, through inquiries conducted during the project. Note also that excess inventory costs are proportional to the number of boxes above the container capacity but do not depend on the number of periods these boxes are kept in the pharmacy before being collected. This way, we privileged disturbance happening at the moment when the box is deposited against risk and disturbance that occur all along the time period when the box is stored out of the container. Every model and method presented subsequently could, however, easily be adapted to this alternative modeling if needed.

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We assume that, at a given period, when a visit is planned, all boxes arrive before the vehicle visits the pharmacies. This assumption relies on the hypothesis that the vehicle visits pharmacies at relatively regular schedules. Hence, random variables can be defined according to that schedule.

The problem can be formulated as a two-stage stochastic optimization problem with recourse. In the first stage, decisions have to be made on the visit dates for each pharmacy and the tours by which the vehicle visits the pharmacies. The actual number of boxes at a pharmacy becomes known when the pharmacy is visited. Therefore, the number of boxes to be collected at each pharmacy is determined in the second stage. Note that for our model, it makes no difference whether the number of boxes at the pharmacies becomes known at once (e.g., assessed via RFID technology), or gradually, going from one pharmacy to the next. This particularity is explained by the fact that in the second stage the whole inventory at each pharmacy is collected.

Let $G = (V, A)$ be a complete graph and $\{1, \ldots, T\}$ a set of discrete time periods. Set $V = \{0, 1, \ldots, n\}$ is the vertex set and A is the arc set. Vertex 0 stands for the vehicle depot. Vertices $i = 1, \ldots, n$ represent the pharmacies. The travel cost on arc $(i, j) \in A$ is denoted c_{ij}; the travel cost matrix is assumed to satisfy the triangle inequality. The fixed cost incurred when carrying out a tour is denoted c_f. Random variables ξ_i^t represent the number of boxes disposed at pharmacy i at time period t. The initial inventory level at pharmacy i is I_i^0. Cost c_i^{thrd} denotes the penalty cost incurred by a visit to pharmacy i, if the threshold inventory level I_i^{thrd} is not reached. Cost $c_i^{\max}$ represents the unitary excess inventory cost incurred by a visit at pharmacy i, if the maximal accepted inventory level $I_i^{\max}$ is exceeded.

The following decision variables are introduced. Binary variable w_i^t is equal to 1 if pharmacy i is visited at time period t, 0 otherwise; z_i^t is the number of boxes picked up during the visit. Flow variable x_{ij}^t is a binary variable set to 1 if pharmacy j is visited immediately after pharmacy i at time period t, 0 otherwise. Variables I_i^t are introduced to compute the inventory level at pharmacy i after arrival of new boxes and before collection of stored boxes at time period t. The nonnegative number of boxes exceeding the maximal accepted inventory is denoted v_i^t; binary variable y_i^t indicates if I_i^{thrd} is reached or not.

Variables x_{ij}^t and w_i^t determine the first-stage decisions, whereas variables z_i^t determine the second-stage decision. Table 1 summarizes the general notation for the SCMIRP.

Using this notation, the SCMIRP can be formulated as follows.

First stage:

$$\min \sum_{i=0}^{n}\sum_{j=0}^{n}\sum_{t=1}^{T} c_{ij}x_{ij}^t + \sum_{j=1}^{n}\sum_{t=1}^{T} c_f x_{0j}^t + \mathbf{E}(Q(x,w,\xi)) \quad \text{s.t.} \tag{1}$$

$$\sum_{j=1}^{n} x_{0j}^t \le 1 \quad \forall t = 1, \ldots, T \tag{2}$$

TABLE 1. Notation.

Parameters	
n	Number of pharmacies
T	Number of time periods
ξ_i^t	Number of boxes disposed at pharmacy i at time t (random variable with Poisson distribution of mean λ_{it})
I_i^0	Initial inventory at pharmacy i
$I_i^{\max}$	Maximal accepted inventory at pharmacy i
I_i^{thrd}	Threshold inventory at pharmacy i
c_{ij}	Travel cost between locations i and j
c_f	Fixed vehicle cost
c_i^{thrd}	Penalty cost at pharmacy i
$c_i^{\max}$	Unitary excess inventory cost at pharmacy i
Variables	
w_i^t	1 if pharmacy i is visited at time t, 0 otherwise
z_i^t	Number of boxes picked up at pharmacy i at time t
x_{ij}^t	1 if pharmacy j is visited immediately after pharmacy i at time t, 0 otherwise
I_i^t	Inventory at pharmacy i at time t
v_i^t	Number of boxes exceeding $I_i^{\max}$ at time t
y_i^t	1 if I_i^{thrd} is not reached when visiting pharmacy i at time t, 0 otherwise

$$\sum_{i=0,\, i\neq j}^{n} x_{ij}^t = w_j^t \quad \forall j = 1, \ldots, n;\ t = 1, \ldots, T \tag{3}$$

$$\sum_{j=0,\, j\neq i}^{n} x_{ij}^t = w_i^t \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{4}$$

$$\sum_{i,j\in S} x_{ij}^t \le |S| - 1 \quad \forall S \subseteq \{1, \ldots, n\};\ t = 1, \ldots, T \tag{5}$$

$$x_{ij}^t \in \{0, 1\} \quad \forall i = 0, \ldots, n;\ j = 0, \ldots, n;\ t = 1, \ldots, T \tag{6}$$

$$w_i^t \in \{0, 1\} \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{7}$$

Second stage:

$$Q(x, w, \xi) = \min \sum_{i=1}^{n}\sum_{t=1}^{T} c_i^{\max} v_i^t + \sum_{i=1}^{n}\sum_{t=1}^{T} c_i^{\text{thrd}} y_i^t + \sum_{i=1}^{n} c_i^{\max} I_i^{T+1} \quad \text{s.t.} \tag{8}$$

$$I_i^t = I_i^0 + \sum_{s=1}^{t} \xi_i^s - \sum_{s=1}^{t-1} z_i^s \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{9}$$

$$I_i^{T+1} \ge I_i^T - z_i^T - I_i^{\max} \quad \forall i = 1, \ldots, n \tag{10}$$

$$z_i^t \le I_i^t + M(1 - w_i^t) \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{11}$$

$$z_i^t \ge I_i^t - M(1 - w_i^t) \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{12}$$

$$z_i^t \le M w_i^t \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{13}$$

$$v_i^t \ge z_i^t - I_i^{\max} \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{14}$$

$$z_i^t \ge I_i^{\text{thrd}}(w_i^t - y_i^t) \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{15}$$

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$$I_i^t \geq 0 \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T+1 \tag{16}$$

$$z_i^t \geq 0 \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{17}$$

$$v_i^t \geq 0 \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{18}$$

$$y_i^t \in \{0, 1\} \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{19}$$

The objective function of the first-stage problem is given by (1). It expresses the minimization of the total cost, which comprises travel costs, fixed costs, and expected inventory costs. The expectation is taken with respect to the distribution of the vector ξ of random variables ξ_i^t. The actual inventory costs $Q(x, w, \xi)$ result as the solution of the second-stage problem (8)–(19). In the terminology of stochastic programming, $Q(x, w, \xi)$ is called the recourse function. It expresses the cost of the decisions taken after the realization of ξ.

Constraints (2) state that at most one tour can be carried out per discrete time period. Constraints (3) and (4) guarantee that a pharmacy is only visited when an arc leads to that pharmacy; they also ensure that if the vehicle visits a pharmacy at a discrete time period, it also leaves the pharmacy at the same time period. Constraints (5) are the subtour elimination constraints. Constraints (6) and (7) define the variables of the first-stage problem.

In the second-stage problem, Equation (8) expresses the objective function as the sum of unitary excess inventory costs and penalty costs (variables I_i^{T+1} represent inventory in excess at the end of the horizon). Constraints (9) and (10) define the level of inventory of pharmacy i at time period t and the inventory in excess at the end of the planning horizon, respectively. Constraints (11) and (12) ensure that the whole inventory is collected when pharmacy i is visited. Constraints (13) force the number of collected boxes to zero if a pharmacy is not visited. Constraints (14), with Constraints (18), determine the nonnegative number of boxes that exceed the maximal accepted inventory at pharmacy i. Constraints (15) define y_i^t as equal to 1 if the number of boxes picked up at pharmacy i at time t is lower than the threshold inventory, and 0 otherwise. Constraints (16) to (19) define the variables of the second-stage problem.

4. SOLUTION APPROACHES

Due to the complexity of the problem, an exact solution method cannot be envisaged to solve realistic problem instances. Therefore, we propose two heuristic methods to solve the SCMIRP for infectious medical waste. The first solution approach is a sampling method. This method is based on solving a mixed integer program in which we consider that the probability distribution of ξ has a finite number of realizations called scenarios. The second solution approach is based on an ALNS algorithm and a collection pattern based operator is developed to handle stochasticity.

4.1. Sampling Method

The aspect of stochasticity in our problem is treated here by performing optimization based on a fixed sample of random scenarios (Birge and Louveaux [5]). In this approach, the given distribution of the random vector ξ is approximated by an empirical sample distribution, obtained by drawing N scenarios ξ^ν (ν = 1, …, N)) from the original distribution of ξ. The original distribution is then replaced by a discrete distribution, each of the N scenarios possessing the same probability $1/N$. The expected unitary excess inventory costs and the expected costs of a visit at a pharmacy are then approximated by the sample average:

$$\frac{1}{N} \sum_{\nu=1}^{N} Q(x, w, \xi^{(\nu)}) \tag{20}$$

By defining $I_{it}^{(\nu)}$, $z_{it}^{(\nu)}$, $v_{it}^{(\nu)}$, and $y_{it}^{(\nu)}$ as the instance of the second-stage variables corresponding to scenario ν, we can combine the first-stage model (1)–(7) and the second-stage model (8)–(19) into a single mixed integer linear program as follows:

$$\begin{aligned} \min \sum_{i=0}^{n} \sum_{j=0}^{n} \sum_{t=1}^{T} c_{ij} x_{ij}^t + \sum_{j=1}^{n} \sum_{t=1}^{T} c_f x_{0j}^t \\ + \frac{1}{N} \sum_{\nu=1}^{N} \left(\sum_{i=1}^{n} \sum_{t=1}^{T} c_i^{\max} v_{it}^{(\nu)} + \sum_{i=1}^{n} \sum_{t=1}^{T} c_i^{\text{thrd}} y_{it}^{(\nu)} \right. \\ \left. + \sum_{i=1}^{n} c_i^{\max} I_{i,T+1}^{(\nu)} \right) \end{aligned} \tag{21}$$

s.t.

$$\sum_{j=1}^{n} x_{0j}^t \leq 1 \quad \forall t = 1, \ldots, T \tag{22}$$

$$\sum_{i=0,\ i \neq j}^{n} x_{ij}^t = w_j^t \quad \forall j = 1, \ldots, n;\ t = 1, \ldots, T \tag{23}$$

$$\sum_{j=0,\ j \neq i}^{n} x_{ij}^t = w_i^t \quad \forall i = 1, \ldots, n;\ t = 1, \ldots, T \tag{24}$$

$$\sum_{i,j \in S} x_{ij}^t \leq |S| - 1 \quad \forall S \subseteq \{1, \ldots, n\};\ t = 1, \ldots, T \tag{25}$$

$$\begin{aligned} I_{it}^{(\nu)} = I_{i0}^{(\nu)} + \sum_{s=1}^{t} \xi_{is}^{(\nu)} - \sum_{s=1}^{t-1} z_{is}^{(\nu)} \\ \forall i = 1, \ldots, n;\ t = 1, \ldots, T;\ \nu = 1, \ldots, N \end{aligned} \tag{26}$$

$$I_{i,T+1}^{(\nu)} \geq I_{iT}^{(\nu)} - z_{iT}^{(\nu)} - I_i^{\max} \quad \forall i = 1, \ldots, n;\ \nu = 1, \ldots, N \tag{27}$$

$$\begin{aligned} z_{it}^{(\nu)} \leq I_{it}^{(\nu)} + M(1 - w_i^t) \\ \forall i = 1, \ldots, n;\ t = 1, \ldots, T;\ \nu = 1, \ldots, N \end{aligned} \tag{28}$$

$$\begin{aligned} z_{it}^{(\nu)} \geq I_{it}^{(\nu)} - M(1 - w_i^t) \\ \forall i = 1, \ldots, n;\ t = 1, \ldots, T;\ \nu = 1, \ldots, N \end{aligned} \tag{29}$$

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$$z_{it}^{(\nu)} \le M w_i^t \quad \forall i = 1,\dots,n;\ t = 1,\dots,T;\ \nu = 1,\dots,N \tag{30}$$

$$v_{it}^{(\nu)} \ge z_{it}^{(\nu)} - I_i^{\max} \quad \forall i = 1,\dots,n;\ t = 1,\dots,T;\ \nu = 1,\dots,N \tag{31}$$

$$z_{it}^{(\nu)} \ge I_i^{\text{thrd}}(w_i^t - y_i^t) \quad \forall i = 1,\dots,n;\ t = 1,\dots,T;\ \nu = 1,\dots,N \tag{32}$$

$$x_{ij}^t \in \{0,1\} \quad \forall i = 0,\dots,n;\ j = 0,\dots,n;\ t = 1,\dots,T \tag{33}$$

$$w_i^t \in \{0,1\} \quad \forall i = 1,\dots,n;\ t = 1,\dots,T \tag{34}$$

$$I_{it}^{(\nu)} \ge 0 \quad \forall i = 1,\dots,n;\ t = 1,\dots,T+1;\ \nu = 1,\dots,N \tag{35}$$

$$z_{it}^{(\nu)} \ge 0 \quad \forall i = 1,\dots,n;\ t = 1,\dots,T;\ \nu = 1,\dots,N \tag{36}$$

$$v_{it}^{(\nu)} \ge 0 \quad \forall i = 1,\dots,n;\ t = 1,\dots,T;\ \nu = 1,\dots,N \tag{37}$$

$$y_{it}^{(\nu)} \in \{0,1\} \quad \forall i = 1,\dots,n;\ t = 1,\dots,T;\ \nu = 1,\dots,N \tag{38}$$

4.2. ALNS Algorithm

ALNS (Pisinger and Ropke [26]) is a large neighborhood search based algorithm, where an initial solution is gradually improved by alternately destroying and repairing the solution.

4.2.1. Basic Scheme As an extension of large neighborhood search, ALNS allows multiple destroy and repair mechanisms (multiple neighborhoods) to be used within the same search. Each of the destroy and repair mechanisms is assigned a weight representing its performance during the search, which controls the selection of a particular mechanism. The weights are adjusted dynamically during the search; therefore, mechanisms that perform well have a higher chance of being selected.

The following pseudocode presents the basic scheme of our ALNS algorithm, which is embedded in a multistart procedure.

Solution s^* represents the best solution found during the execution of the algorithm. Initially, s^* refers to a fictitious solution whose value $f(s^*)$ is artificially set to $+\infty$. Each iteration of loop 2–23 corresponds to a new multistart iteration: a random solution s is constructed (line 4), then improved with ALNS (lines 5–19); once improved, s is compared to s^* and kept if better (lines 20 to 22). Multi-start is repeated until a given time limit is exceeded. Details on the way solutions are evaluated, especially with regard to inventory costs, can be found in section 4.2.2.

Initial solutions are generated by randomly assigning collection days to pharmacies. Then, for each day, a traveling salesman problem (TSP) is solved with the Lin–Kernighan heuristic (Lin and Kernighan [20]). The Lin–Kernighan heuristic is a generalization of 2-opt and 3-opt, swapping pairs of subtours to create a new tour.

Algorithm 1 ALNS algorithm in a multi-start procedure

```
1: f(s*) ← +∞
2: repeat
3:    Initialize probabilities for perturbation
4:    s ← initial solution
5:    s ← local search(s)
6:    repeat
7:       STOP = false
8:       repeat
9:          Select destroy mechanism
10:         s' ← perturbation(s) (destroy mechanism)
11:         s' ← local search(s') (repair mechanism)
12:      until Improvement or fail
13:      if f(s') < f(s) then
14:         s ← s'
15:         Update probabilities for perturbation
16:      else
17:         STOP = true
18:      end if
19:   until STOP
20:   if f(s) < f(s*) then
21:      s* ← s
22:   end if
23: until Time limit
```

The ALNS algorithm relies on four different destroy operators and a single-large neighborhood local search mechanism. These operators and the way they are combined is detailed below.

4.2.2. Destroy and Repair Mechanisms We explore four different diversification mechanisms in the ALNS, which are selected by a roulette wheel method based on their success during the search. At the beginning, each destroy mechanism has the same selection probability, which is increased by 1.5% if the particular operator leads to an improvement of the current solution (probabilities of other operators are decreased by 0.5%). Each perturbation is followed by local search. The perturbated solution s', improved with local search, is only kept when beating current solution s. Otherwise, another destroy operator is tried. When all destroy operators fail, ALNS is stopped.

The four operators are listed below:

- The first operator removes all visits on a random day *d*. For each pharmacy that was visited on day *d*, either day *d-1* or *d+1* is randomly selected, and the pharmacy is added with a best insertion policy (giving the possibility that a new route is created).
- The second destroy operator determines the day *d*, at which least often a tour has been planned. All pharmacies are then inserted on day *d*.
- The third operator temporarily multiplies inventory costs by a factor 3 and applies a local search iteration (see below).
- The fourth destroy operator selects the pharmacy with highest inventory costs and reassigns its visit dates such that expected inventory costs are minimized; this is achieved using the local search operator (see below) and fixing transportation costs to zero.

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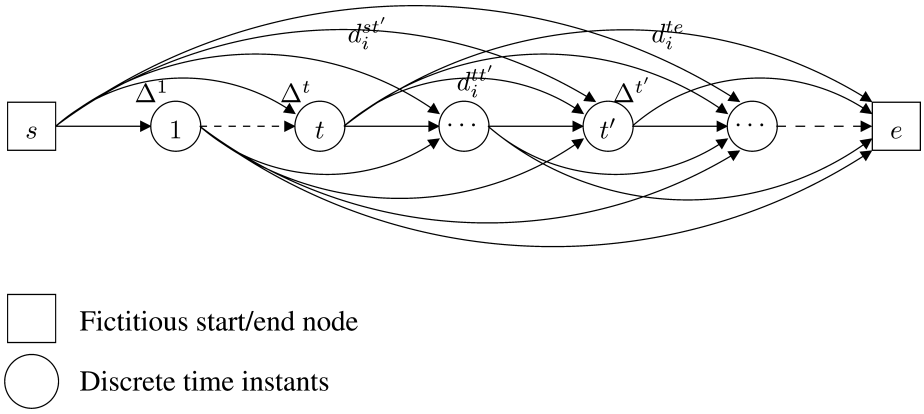


FIG. 3. Shortest path problem for pharmacy i.

The intensification mechanism of our ALNS is inspired by periodic TSPs and is based on collection patterns. A collection pattern is defined as a set of visit dates for a pharmacy. Neighbor solutions are obtained from the current solution by selecting a pharmacy, exchanging the collection pattern for this pharmacy, and reoptimizing vehicle routes with the Lin–Kernighan heuristic, thus defining 2^T neighbor solutions per pharmacy. Local search then consists in selecting the best neighbor solution for pharmacies in turn, until a complete round of the pharmacies shows that current collection patterns are the best ones for all pharmacies.

Figure 3 illustrates the calculation of the best neighbor for a pharmacy i, which is achieved by solving a shortest path problem in the following graph. Each node of the graph represents a discrete time period. Additionally, a fictitious start node and a fictitious end node complete the graph. Visiting a node means that pharmacy i is visited at the corresponding time period (day) t. An arc leads from each node to all of its successors. Paths in the graph then exactly match collection patterns.

The distance (costs) for the shortest path problem is composed of two series of costs. With the traversal of each node is associated a cost Δ^t, which represents the routing cost induced by the visit of pharmacy i on that day. With each arc is associated an inventory cost $d_i^{tt'}$, which computes the expected inventory cost occasioned when planning the next visit of the pharmacy after day t on day t'.

To evaluate quickly costs Δ^t, Δ^t is defined as the pharmacy insertion cost (when pharmacy i is not present in the vehicle tour on day t) or removal saving (when pharmacy i is present). Note that once the best neighbor is found, actual routing costs for the collection pattern are computed using the Lin–Kernighan heuristic, as stated above.

Stochasticity of the SCMIRP is captured in arc costs $d_i^{tt'}$. In the SCMIRP, the number of boxes disposed at a pharmacy cannot be known exactly a priori. Therefore, the number of boxes disposed at pharmacy i by unit of time is expressed by the random variable ξ_i^t. We calculate the estimated inventory costs $d_i^{tt'}$ as the estimated costs of being penalized either because the threshold inventory level is not reached or because the maximal accepted inventory level is exceeded. Given a pharmacy i and two successive visit dates t and t', the estimated inventory costs can be calculated as follows:

$$d_i^{tt'} = c_i^{\max} \sum_{I=I_i^{\max}+1}^{\infty} P(I_i^{tt'} = I) \times (I - I_i^{\max}) + c_i^{\text{thrd}} P(I_i^{tt'} \leq I_i^{\text{thrd}} - 1) \quad (39)$$

$$I_i^{tt'} = \sum_{r=t+1}^{t'} \xi_i^r \quad (40)$$

Equation (39) represents inventory costs, which are composed of unitary excess inventory costs and penalty costs at pharmacy i. To take account of the stochastic data, we calculate the probability of reaching a certain inventory level when pharmacy i is visited. Equation (40) determines the inventory at pharmacy i at time t', if a collection is done at time t.

Regarding inventory costs $d_i^{st'}$ for outgoing arcs of fictitious node s, Equation (40) is replaced with:

$$I_i^{st'} = I_i^0 + \sum_{r=1}^{t'} \xi_i^r \quad (41)$$

computing the accumulated inventory, from the initial inventory I_i^0, at period t'. For the costs d_i^{te}, leading to the fictitious end node, only unitary excess inventory costs are considered. Equation (39) is changed to:

$$d_i^{te} = c_i^{\max} \sum_{I=I_i^{\max}+1}^{\infty} P(I_i^{tT} = I) \times (I - I_i^{\max}) \quad (42)$$

The shortest path is found by applying Dijkstra's algorithm [11] and provides the best collection pattern for pharmacy i, in view of collection patterns for other pharmacies. If the collection pattern represented by the shortest path differs from the current pattern, the solution is updated. Note that at this point, as explained above, Lin–Kernighan heuristic is applied

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to improve routing costs at the days where vehicle routes have changed. Local search then continues for the next pharmacy.

5. COMPUTATIONAL EXPERIMENTS

All experiments are conducted on a PC equipped with an Intel Core2 processor paced at 2.5 GHz and with 3.5 Gb of RAM. Mixed integer linear programs are solved using IBM ILOG CPLEX 11.2 and Concert Technology.

5.1. Test Instances

We tested the proposed solution approaches on several realistic instances from the region of PACA. In this region, pharmacies equipped with RFID technology participate in the project "PAC ID DASRI", with the aim of improving the logistics of infectious medical waste. Information on the number of boxes arriving at each pharmacy and the specification of the threshold inventory level and the maximal accepted inventory level (is) gathered from the participating pharmacies.

To evaluate the proposed solution approaches, numerical experiments are carried out for five small instances and five real-world instances. Concerning the small instances, five pharmacies are considered on a time period of 10 days, each discrete time instant representing 1 day. The fixed vehicle costs are set to 1000, penalty costs (if boxes are collected when the threshold inventory level has not been reached) are 1000, and excess inventory costs are 1000 for each box exceeding the maximal accepted inventory level. The real-world instances contain 30 pharmacies each, with a time horizon of 20 days. For these instances, the fixed vehicle costs are 5000, penalty costs are 1000 and excess inventory costs are 1000 for each box exceeding the maximal accepted inventory level.

We assume that full boxes of infectious medical waste arrive independently at pharmacies according to a Poisson distribution with mean λ_i^t. The estimated inventory costs for pharmacy i are then calculated as follows from Equations (39) and (40).

$$\lambda = \sum_{r=t+1}^{t'} \lambda_i^r \tag{43}$$

$$\begin{aligned} d_i^{tt'} &= c_i^{\max} \sum_{I_i^{tt'}=I_i^{\max}+1}^{\infty} e^{-\lambda} \frac{\lambda^{I_i^{tt'}}}{I_i^{tt'}!} (I_i^{tt'} - I_i^{\max}) \\ &\quad + c_i^{\text{thrd}} \sum_{I_i^{tt'}=0}^{I_i^{\text{thrd}}-1} e^{-\lambda} \frac{\lambda^{I_i^{tt'}}}{I_i^{tt'}!} \end{aligned} \tag{44}$$

$$\begin{aligned} &= c_i^{\max}(\lambda - I_i^{\max} + (I_i^{\max} - \lambda) \sum_{I_i^{tt'}=0}^{I_i^{\max}-1} e^{-\lambda} \frac{\lambda^{I_i^{tt'}}}{I_i^{tt'}!} \\ &\quad + I_i^{\max} e^{-\lambda} \frac{\lambda^{I_i^{\max}}}{I_i^{\max}!}\Big) + c_i^{\text{thrd}} \sum_{I_i^{tt'}=0}^{I_i^{\text{thrd}}-1} e^{-\lambda} \frac{\lambda^{I_i^{tt'}}}{I_i^{tt'}!} \end{aligned}$$

5.2. Evaluation of the Methods

In a first part of our experiments, we evaluate the proposed solution approaches. We proceed with two series of experiments.

First, to assess the performance of our ALNS algorithm, solutions determined by ALNS based on deterministic box arrival data (ALNS deterministic) are compared to the optimal solutions determined with CPLEX (CPLEX realizations). The ALNS algorithm can simply be adapted to a deterministic setting: costs $d_i^{tt'}$ assigned to arc traversals in the collection pattern based local search operator are replaced with real inventory costs. The mixed integer linear program (21)–(38) presented in section 4.1 provides the optimal solution for a given scenario by considering only this scenario in the sample. It can be solved with a cutting-plane method as follows: formulation (21)–(38) is solved without subtour elimination constraints (25), which are added when needed until a feasible solution is found.

This series of experiments are carried out on the set of small instances as the exact solution is out of reach for real-world instances. CPLEX and ALNS are applied to 10 (randomly generated) realizations of box arrivals for each instance. Table 2 presents the average costs and computation times for the 10 realizations for each of the five small instances. Time limit for ALNS is set to 30 s.

The results show that the optimal solution is found by the ALNS algorithm for all 50 test instances, indicating that our algorithm performs very well on small instances.

Then, we compare the quality of the two solution methods introduced in section 4 with stochastic data. Therefore, we oppose the ALNS algorithm (ALNS average) to the sampling method (CPLEX sampling). We complete these comparisons with a third heuristic that computes the optimal solution when box arrival data are set to their average value (CPLEX average). The latter is solved using the sampling formulation with a single-scenario set to average box arrival values. Note that a direct comparison between these approaches is meaningless. Indeed, only ALNS average provides the value of objective function (1). For the sake of the comparison, we evaluate the solution costs *a posteriori* for the two other methods using Equation (44), which allows computing average inventory costs once routes are known. Table 3 reports these results and the computation times. Again these experiments could only be performed on small instances. For the sampling method, 10 randomly generated realizations are used. Mixed integer linear programs are solved with a cutting-plane method as described above. The stopping criterion for ALNS is set to 30 s.

Table 3 shows that our ALNS algorithm can handle stochasticity and determine solutions with a quality significantly better than other methods. CPLEX average is not able to provide solutions that are robust regarding other realizations of the stochastic box arrivals, with gaps up to 43.5%. CPLEX sampling generates solutions that are more robust in terms of unknown realizations of box arrivals but is still incapable of providing robust solutions with respect to

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TABLE 2. Computational results for realizations of small instances (deterministic).

	Routing costs	Inventory costs	Total costs	Computing time
CPLEX realizations				
Instance 1	2863.2	2100.0	4963.2	18 min
Instance 2	2983.8	4200.0	7183.8	45 min
Instance 3	2346.9	700.0	3046.9	10 min
Instance 4	2147.1	1100.0	3247.1	7 min
instance 5	2966.9	2700.0	5666.9	27 min
ALNS deterministic				
Instance 1	2863.2	2100.0	4963.2	30 sec
Instance 2	2983.8	4200.0	7183.8	30 sec
Instance 3	2346.9	700.0	3046.9	30 sec
Instance 4	2147.1	1100.0	3247.1	30 sec
Instance 5	2966.9	2700.0	5666.9	30 sec

every possible realization of box arrivals. Computing times for ALNS are also several order of magnitudes faster than for other methods. It should, however, be noted that the cutting-plane method used for solving the mixed integer linear programs is quite naive and could probably be significantly accelerated using addditional valid inequalities from the TSP polytope.

In what follows, we conduct a sensitivity analysis on routing and inventory costs to show the behavior of the three methods. Initial inventory and routing costs are multiplied by different factors to give more or less importance to each cost component. Table 4 summarizes average costs, average computation times, and average gaps for each cost factors configuration and for each method. Inv. Factor and Rout. Factor represent factors for, respectively, the inventory costs and the routing costs. The stopping criterion for ALNS is set to 30 s.

From Table 4, we can easily notice that computation times increase considerably for CPLEX average and CPLEX sampling when routing costs are high. We can also notice that gaps to ALNS become smaller when routing costs are high. This can be explained by the fact that the stochasticity impacts only inventory costs which play a small role when routing costs are very high.

5.3. *Managerial Implications of RFID*

We now evaluate the performance of different policies for the collection of infectious medical waste. For these experiments, the set of five real-world instances is used. We compare the new policy investigated in this article with three other policies. Remind that the new policy, subsequently called the RFID scenario, refers to the planning of collection tours on the whole time horizon at once, based on stock levels which are known in real time, and statistics on full-box arrival rates. Other policies are:

Current scenario 85%: Pharmacies make a phone call to the local authority when reaching 85% of the maximal accepted inventory. Boxes at a pharmacy are collected the day after having called the local authority. This policy

TABLE 3. Computational results for small instances (stochastic).

	Routing costs	Inventory costs	Total costs	Computing time	Gap to ALNS (%)
CPLEX average					
Instance 1	2497.5	6365.1	8862.6	5 min	17.3
Instance 2	3687.7	10,061.7	13,749.4	30 min	25.8
Instance 3	1269.6	4200.8	5470.4	15 min	0.0
Instance 4	1188.5	9600.4	10,788.9	2 min	43.5
Instance 5	2455.5	7644.9	10,100.4	35 min	3.6
CPLEX sampling					
Instance 1	3637.2	5344.5	8981.7	32 h	18.8
Instance 2	2551.9	8371.0	10,922.9	46 h	0.0
Instance 3	1269.6	4200.8	5470.4	11 h	0.0
Instance 4	2426.7	5945.6	8372.3	2 h	11.4
Instance 5	2455.5	7644.9	10,100.4	72 h	3.6
ALNS average					
Instance 1	2497.7	5059.7	7557.4	30 sec	
Instance 2	2551.9	8371.0	10,922.9	30 sec	
Instance 3	1269.6	4200.8	5470.4	30 sec	
Instance 4	2427.8	5088.7	7516.5	30 sec	
Instance 5	2455.5	7290.9	9746.4	30 sec	

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TABLE 4. Cost analysis for small instances (stochastic).

Method	Rout. Factor	Inv. Factor	Total costs	Routing costs	Inventory costs	Computing time (sec.)	Gap to ALNS
ALNS average	1	10	4236.01	52, 401.68	56, 637.66	30	0.00%
CPLEX average	1	10	3351.52	75, 385.50	78, 737.08	8	44.44%
CPLEX sampling	1	10	4002.17	58, 629.92	62, 632.10	1749	11.77%
ALNS average	1	5	3804.21	26, 467.18	30, 271.36	30	0.00%
CPLEX average	1	5	3351.52	37, 208.96	40, 560.48	9	37.68%
CPLEX sampling	1	5	3787.52	29, 722.70	33, 510.18	2489	10.83%
ALNS average	5	1	10, 014.27	6408.94	16, 423.20	30	0.00%
CPLEX average	5	1	9962.86	7352.36	17, 315.22	11483	5.71%
CPLEX sampling	5	1	10, 049.91	6948.21	16, 998.10	15, 7452	3.99%
ALNS average	10	1	15, 143.84	9400.23	24, 544.08	30	0.00%
CPLEX average	10	1	15, 017.28	12, 844.78	27, 862.04	24, 384	12.98%
CPLEX sampling	10	1	12, 589.62	12, 172.13	24, 761.76	132, 457	0.98%

approximately corresponds to the existing one in the PACA region.

Current scenario threshold: Pharmacies call the local authority when the threshold inventory level is reached. Collection of boxes is performed the same day. This policy is optimized as against the previous one, by assuming a high reactivity of local authorities and a strong intuition (without RFID) or reactivity (with RFID) of the pharmacists.

Rolling horizon scenario: Solutions found by the RFID scenario are revised every day, according to the actual arrivals of data.

All these policies cannot easily be compared on average. Indeed, decisions taken on a given day depend on the actual arrival of boxes on previous days. We thus propose to compare them using a set of 100 realizations of data. To evaluate whether it provides a good approximation of the average behavior of these policies, we first apply this methodology to the RFID scenario.

In Table 5, we compare two measures for ALNS solutions on the five instances. ALNS average refers to the expected costs based on average box arrival rates and corresponds to the solution value returned by the algorithm. ALNS approximated represents the values determined by calculating the actual inventory costs for the 100 realizations based on the solution (collection routes) provided by ALNS. The presented value for each instance includes the mean of the inventory costs for the 100 realizations plus routing costs, which remain unchanged.

Table 5 shows very limited gaps between the two measures. It thus tends to indicate that evaluating the cost of a solution by computing its average cost on 100 realizations gives a good approximation of the expected costs.

Table 6 then compares the four different policies. For all policies, costs are the average values of costs for 100 identical realizations. In the RFID scenario, the time limit is set to 300 s. The solution framework for the Rolling horizon scenario is based on successive calls to the ALNS algorithm, with updated initial inventory levels: every call to ALNS determines the collection route for the following day, which allows, once box arrivals for the next day are known, to determine new inventory levels. Time limit for ALNS is set to 60 s for the first day and to 10 s for the remaining days (where the algorithm is warm-started with the current solution).

It can easily be observed that the total costs for the current scenario 85% are much higher than the costs for the RFID scenario. The advantage of the RFID scenario is that the collector can flexibly combine pharmacies in collection tours to decrease total costs. This is performed by a trade-off between routing costs and inventory costs. For example, it might be preferable to visit a pharmacy which is close-by, even if the threshold inventory level has not been reached. Remember that penalty costs incur when a pharmacy is visited too early (with respect to the threshold inventory) and unitary excess inventory costs incur when a pharmacy is visited too late (with respect to the maximal accepted inventory). In the current scenario 85%, the local authority is not able to prevent visits above the excess inventory level, because a visit is only carried out when a pharmacist makes a call. For this reason, inventory costs are rather high.

In the current scenario threshold, a tour is carried out whenever the local authority receives a call, even if only one pharmacy has to be visited. As the routing of the vehicle can only be optimized on a daily basis, routing costs are very elevated. Due to frequent collections just after the threshold inventory is reached, inventory costs can, however, be kept very low, which leads to total costs comparable to those of the RFID scenario. However, the current scenario threshold requires a strong implication of pharmacists, assumes a high

TABLE 5. Computational results for real-world instances (ALNS).

	ALNS approximated	ALNS average	gap in %
Instance 1	141, 148.4	139, 569.0	1.1
Instance 2	155, 121.7	154, 095.0	0.7
Instance 3	150, 839.2	149, 121.0	1.1
Instance 4	131, 879.4	131, 130.0	0.3
Instance 5	150, 764.0	149, 130.0	1.1

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TABLE 6. Computational results for real-world instances.

	Routing costs	Inventory costs	Total costs
Current scenario 85%			
Instance 1	98, 060.0	156, 870.0	254, 930.0
Instance 2	97, 535.4	177, 960.0	275, 495.4
Instance 3	98, 779.4	185, 810.0	284, 589.4
Instance 4	96, 575.0	140, 610.0	237, 185.0
Instance 5	98, 475.8	179, 700.0	278, 175.8
Mean value	97, 885.1	168, 190.0	266, 075.1
Current scenario threshold			
Instance 1	107, 120.4	31, 830.0	138, 950.4
Instance 2	107, 355.0	39, 390.0	146, 745.0
Instance 3	107, 721.4	37, 510.0	145, 231.4
Instance 4	106, 346.7	25, 810.0	132, 156.7
Instance 5	107, 296.3	38, 270.0	145, 566.3
Mean value	107, 167.9	34, 562.0	141, 729.9
RFID scenario			
Instance 1	47, 868.8	91, 700.0	141, 148.4
Instance 2	53, 114.8	100, 980.0	155, 121.7
Instance 3	42, 700.7	106, 420.0	150, 839.2
Instance 4	42, 630.6	88, 860.0	131, 879.4
Instance 5	47, 939.6	101, 190.0	150, 764.0
Mean value	46, 850.9	97, 830.1	144, 681.0
Rolling horizon scenario			
Instance 1	53, 383.4	71, 650.0	125, 033.4
Instance 2	53, 069.2	83, 540.0	136, 609.2
Instance 3	54, 102.3	81, 920.0	136, 022.3
Instance 4	50, 074.3	68, 310.0	118, 384.3
Instance 5	52, 191.8	82, 080.0	134, 271.8
Mean value	52, 564.2	77, 500.0	130, 064.2

reactivity of the collector and prevents from any long-term planning of his/her activity.

Basically, the current scenario threshold should rather be compared with the rolling horizon scenario, where information on actual arrival of boxes is also exploited to reoptimize collection routes. Results then show that a better balance between routing and inventory costs can be obtained, thus allowing a large improvement of the solution costs.

The conclusion that can be drawn from these results is that the RFID technology can significantly help reducing routing and inventory costs, and that the two aforementioned policies deserve to be considered: the RFID scenario, based on a static planning, and the rolling horizon scenario where updates of the planning are accepted every day.

So as to better compare these two policies, we assessed two additional indicators. These two indicators concern the rolling horizon scenario. The first one states the average number of modifications of the planning over the whole time horizon: one change is counted every time the solution provided by the daily optimization modifies the whole planning. Note that not all of these modifications will be realized in practice as they might concern distant days. The second indicator represents the number of times the daily optimization has a strong impact on the following day, where strong means that the fact that a route is scheduled or not changes. Therefore, the two indicators reflect two extreme cases, on the one hand accounting for each pharmacy that has been rescheduled along the rolling horizon and on the other hand only taking into consideration whether or not a tour has been planned or canceled for the following day. The number of times the collector or the pharmacists are disturbed because of a change of planning (which is quite subjective) certainly stands somewhere between these two extreme measures. Table 7 provides the values of these two indicators for the rolling horizon scenario.

Table 7 exhibits consistent values. On average, 3.1 times within the 20 days a tour is scheduled or canceled for the following day. Thus, the driver for the local authority has to reorganize the planned work load for the next day, causing major inconvenience for him/herself as well as for the

TABLE 7. Average number of changes for the *Rolling horizon scenario*.

Rolling horizon scenario

	Modifications of the whole planning	Strong modifications for the following day
Instance 1	16.7	3.5
Instance 2	16.6	2.8
Instance 3	16.7	3.1
Instance 4	16.5	3.1
Instance 5	16.5	2.8
Mean value	16.6	3.1

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concerned pharmacists. Over the whole time horizon, the initial collection schedule is changed on average 16.6 times.

5.4. Conclusions

The presented computational results enable the following conclusions and implications.

First, the comparison between the optimal solutions for five small instances and the solutions determined by our ALNS algorithm illustrates that our solution approach performs very well and can be applied to generate good approximations of the optima for real-world size problems.

Second, further numerical results on larger instances show that our ALNS algorithm is able to handle stochasticity by computing a reliable expectation of inventory costs.

Third, comparisons with mathematical programming based approaches show that a simple implementation of these methods is not competitive. Quickened solution of the sampling method with larger samples is required to hope beating ALNS.

Fourth, computational results for five real-world instances provide an insight into the different policies which could be pursued by the local authority and the pharmacists. Focusing on the planning for a whole time horizon, the RFID scenario is recommended. It is based on box arrival rate statistics and initial inventory data. A schedule of visit dates and routes for the whole time horizon is then computed, which allows the local authority as well as pharmacists to plan their daily activities and optimize their work schedule. Furthermore, as the routing is optimized by visiting adjacent pharmacies on the same tour, distances and emissions can be lowered. As the planning is performed in advance and based on data gathered via RFID technology, the process can be termed automated and does not require administrative effort of pharmacists or the local authority. This results in less disturbance for all involved parties.

The embedding of the RFID scenario in a rolling horizon framework allows further improvements of all cost components, as the data registered via RFID is used every day to optimize the time schedule for the remaining time horizon. However, this implies that changes in planning are likely to occur and the involved parties cannot rely on the provided schedules. Therefore, flexibility and the potential to adapt to new situations are required, rendering the process of work scheduling less convenient for the pharmacists and the local authority.

6. SUMMARY AND PERSPECTIVES

We presented several solution approaches to optimize the determination of visit dates and the corresponding vehicle routes for infectious medical waste collection based on stochastic input data. We consider social objectives, such as the satisfaction of pharmacists and local authority and the minimization of public health risks. Experimental results show, that by integrating an RFID technology, the planning process can be ameliorated for both, local authority and pharmacists.

The present solution approaches are to be embedded into a decision support system. Together with the decision makers (pharmacists, local authority), the weight for the inventory costs and the threshold and maximal accepted inventory levels can be determined. These values can be chosen independently for each pharmacy, to reflect the inconvenience caused by a visit or excess of inventory.

In the basic model, we do not consider vehicle capacity or tour length constraints, as they were not relevant for our case-study. For the sake of more generality, these constraints could be included in the future. In addition, a further developed version might also consider the inventory level at the collection centers to optimize the distribution of the boxes among the collections centers.

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